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FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
AU-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level audit and accountability policy that a. Addresses	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures at	5	
AU-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level audit and accountability policy that a. Addresses	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
AU-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level audit and accountability policy that a. Addresses	Functional	Subset Of	Continuous Monitoring	MON-01	Mechanisms exist to facilitate the implementation of enterprise-wide monitoring controls.	10	
AU-02	Event Logging	a. Identify the types of events that the system is capable of logging in support of the audit function: [Assignment: organization-defined event types that the system is capable of logging];b. Coordinate the event logging function with other	Functional	Intersects With	Security Event Monitoring	MON-01.8	Mechanisms exist to review event logs on an ongoing basis and escalate incidents in accordance with	5	
AU-02	Event Logging	a. Identify the types of events that the system is capable of logging in support of the audit function: [Assignment: organization-defined event types that the system is capable of logging];b. Coordinate the event logging function with other	Functional	Intersects With	Centralized Collection of Security Event Logs	MON-02	Mechanisms exist to utilize a Security Incident Event Manager (SIEM), or similar automated tool, to support the centralized collection of security-	5	
AU-03	Content of Audit Records	Ensure that audit records contain information that establishes the following:a. What type of event occurred;b. When the event occurred;c. Where the event occurred;d. Source of the event;e. Outcome of the event; andf. Identity of any individuals.	Functional	Equal	Content of Event Logs	MON-03	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) to produce most law enforcement industry	10	
AU-04	Audit Log Storage Capacity	Allocate audit log storage capacity to accommodate [Assignment: organization-defined audit log retention requirements].	Functional	Equal	Event Log Storage Capacity	MON-04	Mechanisms exist to allocate and proactively manage sufficient event log storage capacity to reduce the likelihood of such capacity being	10	
AU-04(01)	Audit Log Storage Capacity Transfer to Alternate Storage	Transfer audit logs [Assignment: organization-defined frequency] to a different system, system component, or media other than the system or system component conducting the logging	Functional	Intersects With	Event Log Backup on Separate Physical Systems / Components	MON-08.1	Mechanisms exist to back up event logs onto a physically different system or system component than the Security Incident Event Manager	5	
AU-05	Response to Audit Logging Process Failures	a. Alert [Assignment: organization-defined personnel or roles] within [Assignment: organization-defined time period] in the event of an audit logging process failure; andb. Take the following additional actions: [Assignment: organization-defined	Functional	Equal	Response To Event Log Processing Failures	MON-05	Mechanisms exist to alert appropriate personnel in the event of a log processing failure and take actions to remedy the disruption.	10	
AU-06	Audit Record Review, Analysis, and Reporting	a. Review and analyze system audit records [Assignment: organization-defined frequency] for indications of [Assignment: organization-defined inappropriate or unusual activity] and the potential impact of the inappropriate or unusual activity;b.	Functional	Intersects With	Centralized Collection of Security Event Logs	MON-02	Mechanisms exist to utilize a Security Incident Event Manager (SIEM), or similar automated tool, to support the centralized collection of security-	5	
AU-06	Audit Record Review, Analysis, and Reporting	a. Review and analyze system audit records [Assignment: organization-defined frequency] for indications of [Assignment: organization-defined inappropriate or unusual activity] and the potential impact of the inappropriate or unusual activity;b.	Functional	Intersects With	Audit Level Adjustments	MON-02.6	Mechanisms exist to adjust the level of audit review, analysis and reporting based on evolving threat information from law enforcement industry	5	
AU-08	Time Stamps	a. Use internal system clocks to generate time stamps for audit records; andb. Record time stamps for audit records that meet [Assignment: organization-defined granularity of time measurement] and that use Coordinated Universal Time, have	Functional	Intersects With	Clock Synchronization	SEA-20	Mechanisms exist to utilize time-synchronization technology to synchronize all critical system clocks.	5	
AU-08	Time Stamps	a. Use internal system clocks to generate time stamps for audit records; andb. Record time stamps for audit records that meet [Assignment: organization-defined granularity of time measurement] and that use Coordinated Universal Time, have	Functional	Intersects With	Time Stamps	MON-07	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) to use an	5	
AU-09	Protection of Audit Information	a. Protect audit information and audit logging tools from unauthorized access, modification, and deletion; andb. Alert [Assignment: organization-defined personnel or roles] upon detection of unauthorized access, modification, or deletion of	Functional	Equal	Protection of Event Logs	MON-08	Mechanisms exist to protect event logs and audit tools from unauthorized access, modification and deletion	10	
AU-11	Audit Record Retention	Retain audit records for [Assignment: organization-defined time period consistent with records retention policy] to provide support for after-the-fact investigations of incidents and to meet regulatory and organizational information retention	Functional	Equal	Event Log Retention	MON-10	Mechanisms exist to retain event logs for a time period consistent with records retention requirements to provide support for after-the-fact	10	
AU-12	Audit Record Generation	a. Provide audit record generation capability for the event types the system is capable of auditing as defined in AU-2a on [Assignment: organization-defined system components];b. Allow [Assignment: organization-defined personnel or roles] to	Functional	Intersects With	Monitoring Reporting	MON-06	Mechanisms exist to provide an event log report generation capability to aid in detecting and assessing anomalous activities	5	
CA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level assessment, authorization and monitoring	Functional	Subset Of	Information Assurance (IA) Operations	IAO-01	Mechanisms exist to facilitate the implementation of security, compliance and resilience assessment and authorization	10	
CA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level assessment, authorization and monitoring	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures at	5	
CA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level assessment, authorization and monitoring	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
CA-02	Control Assessments	a. Select the appropriate assessor or assessment team for the type of assessment to be conducted;b. Develop a control assessment plan that describes the scope of the assessment including:1. Controls and control enhancements under	Functional	Intersects With	Functional Review Of Security, Compliance & Resilience Controls	CPL-03.2	Mechanisms exist to regularly review Technology Assets, Applications and/or Services (TAAS) for adherence	5	
CA-02	Control Assessments	a. Select the appropriate assessor or assessment team for the type of assessment to be conducted;b. Develop a control assessment plan that describes the scope of the assessment including:1. Controls and control enhancements under	Functional	Intersects With	Technical Verification	IAO-06	Mechanisms exist to perform Information Assurance Program (IAP) activities to evaluate the design, implementation and effectiveness of	5	
CA-02	Control Assessments	a. Select the appropriate assessor or assessment team for the type of assessment to be conducted;b. Develop a control assessment plan that describes the scope of the assessment including:1. Controls and control enhancements under	Functional	Intersects With	Security, Compliance & Resilience In Project Management	PRM-04	Mechanisms exist to assess security, compliance and resilience controls as part of Technology Assets, Applications and/or Services (TAAS)	5	
CA-02	Control Assessments	a. Select the appropriate assessor or assessment team for the type of assessment to be conducted;b. Develop a control assessment plan that describes the scope of the assessment including:1. Controls and control enhancements under	Functional	Intersects With	Assessments	IAO-02	Mechanisms exist to formally assess the security, compliance and resilience controls in Technology Assets, Applications and/or Services	5	
CA-02	Control Assessments	a. Select the appropriate assessor or assessment team for the type of assessment to be conducted;b. Develop a control assessment plan that describes the scope of the assessment including:1. Controls and control enhancements under	Functional	Intersects With	Control Conformity Monitoring	CPL-03	Mechanisms exist to validate that Technology Assets, Applications, Services and/or Data (TAASD)	5	
CA-03	Information Exchange	a. Approve and manage the exchange of information between the system and other systems using [Selection (one or more): interconnection security agreements; information exchange security agreements; memoranda of understanding or	Functional	Intersects With	Interconnection Security Agreements (ISAs)	NET-05	Mechanisms exist to authorize connections from systems to other systems using Interconnection Security Agreements (ISAs), or similar	5	

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
CA-05	Plan of Action and Milestones	a. Develop a plan of action and milestones for the system to document the planned remediation actions of the organization to correct weaknesses or deficiencies noted during the assessment of the controls and to reduce or eliminate known weaknesses.	Functional	Intersects With	Capabilities Deficiency Tracking	IAO-05	Mechanisms exist to govern identified deficiencies (e.g., Plan of Action and Milestones (POA&M) or similar methodology) that formally track and document the remediation of weaknesses.	5	
CA-06	Authorization	a. Assign a senior official as the authorizing official for the system; b. Assign a senior official as the authorizing official for common controls available for inheritance by organizational systems; c. Ensure that the authorizing official for the system is not the same as the authorizing official for the system.	Functional	Equal	Security Authorization	IAO-07	Mechanisms exist to ensure Technology Assets, Applications and/or Services (TAAS) are officially authorized and are not used in a manner that is inconsistent with the organization's executive policy.	10	
CA-07	Continuous Monitoring	Develop a system-level continuous monitoring strategy and implement continuous monitoring in accordance with the organization-level continuous monitoring strategy that includes: a. Establishing the following system-level metrics to monitor: b. Compliance monitoring; and c. Change monitoring.	Functional	Intersects With	Security, Compliance & Resilience Controls Oversight	CPL-02	Mechanisms exist to provide a security, compliance and resilience controls oversight function that reports to the organization's executive.	5	
CA-07(04)	Continuous Monitoring Risk Monitoring	Ensure risk monitoring is an integral part of the continuous monitoring strategy that includes the following: a. Effectiveness monitoring; b. Compliance monitoring; and c. Change monitoring.	Functional	Equal	Risk Monitoring	RSK-11	Mechanisms exist to ensure risk monitoring as an integral part of the continuous monitoring strategy that includes monitoring the effectiveness of controls.	10	
CA-09	Internal System Connections	a. Authorize internal connections of [Assignment: organization-defined system components or classes of components] to the system; b. Document, for each internal connection, the interface characteristics, security and privacy requirements.	Functional	Equal	Internal System Connections	NET-05.2	Mechanisms exist to control internal system connections through authorizing internal connections of systems and documenting, for each system, the interface characteristics, security and privacy requirements.	10	
CM-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] configuration management policy that: a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] configuration management policy that: a.	Functional	Subset Of	Configuration Management Program	CFG-01	Mechanisms exist to facilitate the implementation of configuration management controls.	10	
CM-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] configuration management policy that: a.	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures necessary for secure, compliant and resilient systems.	5	
CM-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] configuration management policy that: a.	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient systems.	5	
CM-02	Baseline Configuration	a. Develop, document, and maintain under configuration control, a current baseline configuration of the system; and b. Review and update the baseline configuration of the system: 1. [Assignment: organization-defined frequency]; 2. When [Assignment: organization-defined conditions] occur.	Functional	Intersects With	Reviews & Updates	CFG-02.1	Mechanisms exist to review and update baseline configurations: (1) At least annually; (2) When required due to: a. or b. [Assignment: organization-defined conditions] occur.	5	
CM-02	Baseline Configuration	a. Develop, document, and maintain under configuration control, a current baseline configuration of the system; and b. Review and update the baseline configuration of the system: 1. [Assignment: organization-defined frequency]; 2. When [Assignment: organization-defined conditions] occur.	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS).	5	
CM-04	Impact Analyses	Analyze changes to the system to determine potential security and privacy impacts prior to change implementation.	Functional	Equal	Security Impact Analysis for Changes	CHG-03	Mechanisms exist to analyze proposed changes for potential security impacts, prior to the implementation of the change.	10	
CM-05	Access Restrictions for Change	Define, document, approve, and enforce physical and logical access restrictions associated with changes to the system.	Functional	Intersects With	Governing Access Restriction for Change	END-03.2	Mechanisms exist to define, document, approve and enforce access restrictions associated with changes to the system.	5	
CM-05	Access Restrictions for Change	Define, document, approve, and enforce physical and logical access restrictions associated with changes to the system.	Functional	Intersects With	Access Restriction For Change	CHG-04	Mechanisms exist to enforce configuration restrictions in an effort to restrict the ability of users to conduct unauthorized changes.	5	
CM-06	Configuration Settings	a. Establish and document configuration settings for components employed within the system that reflect the most restrictive mode consistent with operational requirements using [Assignment: organization-defined common secure configuration settings]; b. Review and update the configuration settings of the system: 1. [Assignment: organization-defined frequency]; 2. When [Assignment: organization-defined conditions] occur.	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS).	5	
CM-06	Configuration Settings	a. Establish and document configuration settings for components employed within the system that reflect the most restrictive mode consistent with operational requirements using [Assignment: organization-defined common secure configuration settings]; b. Review and update the configuration settings of the system: 1. [Assignment: organization-defined frequency]; 2. When [Assignment: organization-defined conditions] occur.	Functional	Intersects With	Approved Configuration Deviations	CFG-02.7	Mechanisms exist to document, assess risk and approve or deny deviations to standardized configurations.	5	
CM-07	Least Functionality	a. Configure the system to provide only [Assignment: organization-defined mission essential capabilities]; and b. Prohibit or restrict the use of the following functions, ports, protocols, software, and/or services: [Assignment: organization-defined non-essential capabilities].	Functional	Equal	Least Functionality	CFG-03	Mechanisms exist to configure systems to provide only essential capabilities by specifically prohibiting or restricting the use of ports, protocols, software, and/or services.	10	
CM-08	System Component Inventory	a. Develop and document an inventory of system components that: 1. Accurately reflects the system; 2. Includes all components within the system; 3. Does not include duplicate accounting of components or components assigned to any other system.	Functional	Intersects With	Asset Inventories	AST-02	Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data (TAASAPSD).	5	
CM-08	System Component Inventory	a. Develop and document an inventory of system components that: 1. Accurately reflects the system; 2. Includes all components within the system; 3. Does not include duplicate accounting of components or components assigned to any other system.	Functional	Intersects With	Component Duplication Avoidance	AST-02.3	Mechanisms exist to establish and maintain an authoritative source and repository to provide a trusted source and accountability for approved and documented configurations.	5	
CM-10	Software Usage Restrictions	a. Use software and associated documentation in accordance with contract agreements and copyright laws; b. Track the use of software and associated documentation protected by quantity licenses to control copying and distribution; and c. Establish [Assignment: organization-defined policies] governing the installation of software by users; b. Enforce software installation policies through the following methods: [Assignment: organization-defined methods]; and c. Monitor [Assignment: organization-defined metrics] to ensure compliance with software usage restrictions.	Functional	Equal	Software Usage Restrictions	CFG-04	Mechanisms exist to enforce software usage restrictions to comply with applicable contract agreements and copyright laws.	10	
CM-11	User-installed Software	a. Establish [Assignment: organization-defined policies] governing the installation of software by users; b. Enforce software installation policies through the following methods: [Assignment: organization-defined methods]; and c. Monitor [Assignment: organization-defined metrics] to ensure compliance with software usage restrictions.	Functional	Intersects With	User-Installed Software	CFG-05	Mechanisms exist to restrict the ability of non-privileged users to install unauthorized software.	5	
CP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] contingency planning policy that: a. Addresses [Assignment: organization-defined contingency planning policy] that: a.	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures necessary for secure, compliant and resilient systems.	5	
CP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] contingency planning policy that: a. Addresses [Assignment: organization-defined contingency planning policy] that: a.	Functional	Subset Of	Business Continuity Management System (BCMS)	BCD-01	Mechanisms exist to facilitate the implementation of contingency planning controls to help ensure the organization's ability to maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient systems.	10	
CP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] contingency planning policy that: a. Addresses [Assignment: organization-defined contingency planning policy] that: a.	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient systems.	5	
CP-02	Contingency Plan	a. Develop a contingency plan for the system that: 1. Identifies essential mission and business functions and associated contingency requirements; 2. Provides recovery objectives, restoration priorities, and metrics; 3. Addresses contingency planning controls to help ensure the organization's ability to maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient systems.	Functional	Subset Of	Business Continuity Management System (BCMS)	BCD-01	Mechanisms exist to facilitate the implementation of contingency planning controls to help ensure the organization's ability to maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient systems.	10	
CP-02	Contingency Plan	a. Develop a contingency plan for the system that: 1. Identifies essential mission and business functions and associated contingency requirements; 2. Provides recovery objectives, restoration priorities, and metrics; 3. Addresses contingency planning controls to help ensure the organization's ability to maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient systems.	Functional	Intersects With	Ongoing Contingency Planning	BCD-06	Mechanisms exist to update contingency plans due to changes affecting: (1) People (e.g., personnel changes); (2) Processes (e.g., changes to procedures); (3) Technology Assets, Applications, Services and/or Data (TAASAPSD); and (4) Other factors.	5	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm>

Focal Document: **NIST SP 800 - R2 R3 - Guide to Operational Technology (OT) Security - Low OT Over**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3-low.1>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
CP-03	Contingency Training	a. Provide contingency training to system users consistent with assigned roles and responsibilities:1. Within [Assignment: organization-defined time period] of assuming a contingency role or responsibility:2. When required by system changes:	Functional	Equal	Contingency Training	BCD-03	Mechanisms exist to adequately train contingency personnel and applicable stakeholders in their contingency roles and	10	
CP-04	Contingency Plan Testing	a. Test the contingency plan for the system [Assignment: organization-defined frequency] using the following tests to determine the effectiveness of the plan and the readiness to execute the plan: [Assignment: organization-defined tests] b.	Functional	Intersects With	Contingency Plan Root Cause Analysis (RCA) & Lessons Learned	BCD-05	Mechanisms exist to conduct a Root Cause Analysis (RCA) and "lessons learned" activity every time the contingency plan is activated	5	
CP-04	Contingency Plan Testing	a. Test the contingency plan for the system [Assignment: organization-defined frequency] using the following tests to determine the effectiveness of the plan and the readiness to execute the plan: [Assignment: organization-defined tests] b.	Functional	Intersects With	Contingency Plan Testing & Exercises	BCD-04	Mechanisms exist to conduct tests and/or exercises to evaluate the contingency plan's effectiveness and the organization's readiness to	5	
CP-09	System Backup	a. Conduct backups of user-level information contained in [Assignment: organization-defined system components] [Assignment: organization-defined frequency consistent with recovery time and recovery point objectives] b. Conduct	Functional	Intersects With	Data Backups	BCD-11	Mechanisms exist to create recurring backups of data, software and/or system images, as well as verify the integrity of these backups to ensure	5	
CP-10	System Recovery and Reconstitution	Provide for the recovery and reconstitution of the system to a known state within [Assignment: organization-defined time period consistent with recovery time and recovery point objectives] after a disruption, compromise, or failure.	Functional	Intersects With	Technology Assets, Applications and/or Services (TAAS) Recovery &	BCD-12	Mechanisms exist to ensure the secure recovery and reconstitution of Technology Assets, Applications	5	
CP-10	System Recovery and Reconstitution	Provide for the recovery and reconstitution of the system to a known state within [Assignment: organization-defined time period consistent with recovery time and recovery point objectives] after a disruption, compromise, or failure.	Functional	Intersects With	Business Continuity Management System (BCMS)	BCD-01	Mechanisms exist to facilitate the implementation of contingency planning controls to help ensure resilient Technology Assets	5	
CP-10	System Recovery and Reconstitution	Provide for the recovery and reconstitution of the system to a known state within [Assignment: organization-defined time period consistent with recovery time and recovery point objectives] after a disruption, compromise, or failure.	Functional	Intersects With	Recovery Time / Point Objectives (RTO / RPO)	BCD-01.4	Mechanisms exist to facilitate recovery operations in accordance with Recovery Time Objectives (RTOs) and Recovery Point Objectives	5	
CP-12	Safe Mode	When [Assignment: organization-defined conditions] are detected, enter a safe mode of operation with [Assignment: organization-defined restrictions of safe mode of operation].	Functional	Intersects With	Fail Secure	SEA-07.2	Mechanisms exist to enable systems to fail to an organization-defined known-state for types of failures, preserving system state information	5	
IA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] identification and authentication policy that: a.	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures, at	5	
IA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] identification and authentication policy that: a.	Functional	Subset Of	Identity & Access Management (IAM)	IAC-01	Mechanisms exist to facilitate the implementation of identification and access management controls.	10	
IA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] identification and authentication policy that: a.	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
IA-02	Identification and Authentication (organizational Users)	Uniquely identify and authenticate organizational users and associate that unique identification with processes acting on behalf of those users.	Functional	Equal	Identification & Authentication for Organizational Users	IAC-02	Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) organizational users and processes acting on behalf of	10	
IA-02(01)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Multi-Factor Authentication (MFA)	IAC-06	Automated mechanisms exist to enforce Multi-Factor Authentication (MFA) for:	5	
IA-02(01)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Local Access to Privileged Accounts	IAC-06.3	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate local access for privileged accounts.	5	
IA-02(01)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Information Assurance Enabled Products	TDA-02.2	Mechanisms exist to limit the use of commercially-provided Information Assurance (IA) and IA-enabled IT products to those products that have	5	
IA-02(01)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Out-of-Band Multi-Factor Authentication	IAC-06.4	Mechanisms exist to implement Multi-Factor Authentication (MFA) for access to privileged and non-privileged accounts such that one of	5	
IA-02(01)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Network Access to Privileged Accounts	IAC-06.1	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate network access for privileged accounts.	5	
IA-02(01)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Network Access to Non-Privileged Accounts	IAC-06.2	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate network access for non-privileged accounts.	5	
IA-02(01)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Hardware Token-Based Authentication	IAC-10.7	Automated mechanisms exist to ensure organization-defined token quality requirements are satisfied for hardware token-based	5	
IA-02(02)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Information Assurance Enabled Products	TDA-02.2	Mechanisms exist to limit the use of commercially-provided Information Assurance (IA) and IA-enabled IT products to those products that have	5	
IA-02(02)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Network Access to Non-Privileged Accounts	IAC-06.2	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate network access for non-privileged accounts.	5	
IA-02(02)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Out-of-Band Multi-Factor Authentication	IAC-06.4	Mechanisms exist to implement Multi-Factor Authentication (MFA) for access to privileged and non-privileged accounts such that one of	5	
IA-02(02)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Hardware Token-Based Authentication	IAC-10.7	Automated mechanisms exist to ensure organization-defined token quality requirements are satisfied for hardware token-based	5	
IA-02(02)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Network Access to Privileged Accounts	IAC-06.1	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate network access for privileged accounts.	5	
IA-02(02)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Multi-Factor Authentication (MFA)	IAC-06	Automated mechanisms exist to enforce Multi-Factor Authentication (MFA) for:	5	

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
IA-02(02)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Local Access to Privileged Accounts	IAC-06.3	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate local access for privileged accounts.	5	
IA-02(08)	Identification and Authentication (organizational Users) Access to Accounts	Implement replay-resistant authentication mechanisms for access to [Selection (one or more): privileged accounts; non-privileged accounts].	Functional	Equal	Replay-Resistant Authentication	IAC-02.2	Automated mechanisms exist to employ replay-resistant authentication.	10	
IA-02(12)	Identification and Authentication (organizational Users) Acceptance of PIV	Accept and electronically verify Personal Identity Verification-compliant credentials.	Functional	Intersects With	Acceptance of PIV Credentials	IAC-02.3	Mechanisms exist to accept and electronically verify organizational Personal Identity Verification (PIV) credentials.	5	
IA-03	Device Identification and Authentication	Uniquely identify and authenticate [Assignment: organization-defined devices and/or types of devices] before establishing a [Selection (one or more): local; remote; network] connection.	Functional	Intersects With	Identification & Authentication for Devices	IAC-04	Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) devices before establishing a connection using	5	
IA-04	Identifier Management	Manage system identifiers by:a. Receiving authorization from [Assignment: organization-defined personnel or roles] to assign an individual, group, role, service, or device identifier;b. Selecting an identifier that identifies an individual, group, role	Functional	Intersects With	Authenticate, Authorize and Audit (AAA)	IAC-01.2	Mechanisms exist to strictly govern the use of Authenticate, Authorize and Audit (AAA) solutions, both on-premises and those hosted by an	5	
IA-04	Identifier Management	Manage system identifiers by:a. Receiving authorization from [Assignment: organization-defined personnel or roles] to assign an individual, group, role, service, or device identifier;b. Selecting an identifier that identifies an individual, group, role	Functional	Intersects With	Identifier Management (User Names)	IAC-09	Mechanisms exist to govern naming standards for usernames and Technology Assets, Applications and/or Services (TAAS)	5	
IA-05	Authenticator Management	Manage system authenticators by:a. Verifying, as part of the initial authenticator distribution, the identity of the individual, group, role, service, or device receiving the authenticator;b. Establishing initial authenticator content for any authenticators	Functional	Intersects With	Authenticator Management	IAC-10	Mechanisms exist to: (1) Securely manage authenticators for users and devices; and	5	
IA-05	Authenticator Management	Manage system authenticators by:a. Verifying, as part of the initial authenticator distribution, the identity of the individual, group, role, service, or device receiving the authenticator;b. Establishing initial authenticator content for any authenticators	Functional	Intersects With	Default Authenticators	IAC-10.8	Mechanisms exist to ensure default authenticators are changed as part of account creation or system installation	5	
IA-05(01)	Authenticator Management Password-based Authentication	For password-based authentication:a. Maintain a list of commonly-used, expected, or compromised passwords and update the list [Assignment: organization-defined frequency] and when organizational passwords are suspected to have	Functional	Intersects With	Automated Support For Password Strength	IAC-10.4	Automated mechanisms exist to determine if password authenticators are sufficiently strong enough to satisfy organization-defined password	5	
IA-05(01)	Authenticator Management Password-based Authentication	For password-based authentication:a. Maintain a list of commonly-used, expected, or compromised passwords and update the list [Assignment: organization-defined frequency] and when organizational passwords are suspected to have	Functional	Intersects With	Password-Based Authentication	IAC-10.1	Mechanisms exist to enforce complexity, length and lifespan considerations to ensure strong criteria for password-based	5	
IA-05(01)	Authenticator Management Password-based Authentication	For password-based authentication:a. Maintain a list of commonly-used, expected, or compromised passwords and update the list [Assignment: organization-defined frequency] and when organizational passwords are suspected to have	Functional	Intersects With	Authenticator Management	IAC-10	Mechanisms exist to: (1) Securely manage authenticators for users and devices; and	5	
IA-06	Authentication Feedback	Obscure feedback of authentication information during the authentication process to protect the information from possible exploitation and use by unauthorized individuals.	Functional	Equal	Authenticator Feedback	IAC-11	Mechanisms exist to obscure the feedback of authentication information during the authentication process to protect the information	10	
IA-07	Cryptographic Module Authentication	Implement mechanisms for authentication to a cryptographic module that meet the requirements of applicable laws, executive orders, directives, policies, regulations, standards, and guidelines for such authentication	Functional	Intersects With	Cryptographic Module Authentication	IAC-12	Mechanisms exist to ensure cryptographic modules adhere to applicable statutory, regulatory and contractual requirements for security	5	
IA-07	Cryptographic Module Authentication	Implement mechanisms for authentication to a cryptographic module that meet the requirements of applicable laws, executive orders, directives, policies, regulations, standards, and guidelines for such authentication	Functional	Intersects With	Automated Authentication Through Cryptographic	CRY-02	Automated mechanisms exist to enable systems to authenticate to a cryptographic module.	5	
IA-08	Identification and Authentication (non-organizational Users)	Uniquely identify and authenticate non-organizational users or processes acting on behalf of non-organizational users.	Functional	Equal	Identification & Authentication for Non-Organizational Users	IAC-03	Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) third-party users and processes that provide services to the	10	
IA-08(01)	Identification and Authentication (non-organizational Users) Acceptance of PIV	Accept and electronically verify Personal Identity Verification-compliant credentials from other federal agencies.	Functional	Equal	Acceptance of PIV Credentials from Other Organizations	IAC-03.1	Mechanisms exist to accept and electronically verify Personal Identity Verification (PIV) credentials from third-parties	10	
IA-08(02)	Identification and Authentication (non-organizational Users) Acceptance of PIV	a. Accept only external authenticators that are NIST-compliant; andb. Document and maintain a list of accepted external authenticators.	Functional	Equal	Acceptance of Third-Party Credentials	IAC-03.2	Automated mechanisms exist to accept Federal Identity, Credential and Access Management (FICAM)-approved third-party credentials	10	
IA-08(04)	Identification and Authentication (non-organizational Users) Use of Defined	Conform to the following profiles for identity management [Assignment: organization-defined identity management profiles].	Functional	Equal	Use of FICAM-Issued Profiles	IAC-03.3	Mechanisms exist to conform systems to Federal Identity, Credential and Access Management (FICAM)-issued profiles	10	
IA-11	Re-authentication	Require users to re-authenticate when [Assignment: organization-defined circumstances or situations requiring re-authentication].	Functional	Equal	Re-Authentication	IAC-14	Mechanisms exist to force users and devices to re-authenticate according to organization-defined circumstances that necessitate re-	10	
IR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] incident response policy that:a. Addresses	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
IR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] incident response policy that:a. Addresses	Functional	Subset Of	Incident Response Operations	IRO-01	Mechanisms exist to implement and govern processes and documentation to facilitate an organization-wide response capability for cybersecurity	10	
IR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] incident response policy that:a. Addresses	Functional	Intersects With	IRP Update	IRO-04.2	Mechanisms exist to regularly review and modify incident response practices to incorporate lessons learned, business process changes	5	
IR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] incident response policy that:a. Addresses	Functional	Intersects With	Root Cause Analysis (RCA) & Lessons Learned	IRO-13	Mechanisms exist to incorporate lessons learned from analyzing and resolving cybersecurity and data protection incidents to reduce the	5	
IR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] incident response policy that:a. Addresses	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRIP), including policies, standards and procedures, at	5	
IR-02	Incident Response Training	a. Provide incident response training to system users consistent with assigned roles and responsibilities:1. Within [Assignment: organization-defined time period] of assuming an incident response role or responsibility or acquiring system	Functional	Intersects With	Incident Response Training	IRO-05	Mechanisms exist to train personnel in their incident response roles and responsibilities.	5	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Over**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3-low.1>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
IR-04	Incident Handling	a. Implement an incident handling capability for incidents that is consistent with the incident response plan and includes preparation, detection and analysis, containment, eradication, and recovery; b. Coordinate incident handling activities with	Functional	Equal	Incident Handling	IRO-02	Mechanisms exist to cover: (1) Preparation; (2) Automated event detection or manual incident report intake;	10	
IR-05	Incident Monitoring	Track and document incidents.	Functional	Equal	Situational Awareness For Incidents	IRO-09	Mechanisms exist to document, monitor and report the status of cybersecurity and data protection incidents to internal stakeholders all	10	
IR-06	Incident Reporting	a. Require personnel to report suspected incidents to the organizational incident response capability within [Assignment: organization-defined time period]; and b. Report incident information to [Assignment: organization-defined authorities];	Functional	Intersects With	Incident Stakeholder Reporting	IRO-10	Mechanisms exist to timely-report incidents to applicable: (1) Internal stakeholders; (2) Affected clients & third-parties;	5	
IR-06	Incident Reporting	a. Require personnel to report suspected incidents to the organizational incident response capability within [Assignment: organization-defined time period]; and b. Report incident information to [Assignment: organization-defined authorities];	Functional	Intersects With	Regulatory & Law Enforcement Contacts	IRO-14	Mechanisms exist to maintain incident response contacts with applicable regulatory and law enforcement agencies	5	
IR-06	Incident Reporting	a. Require personnel to report suspected incidents to the organizational incident response capability within [Assignment: organization-defined time period]; and b. Report incident information to [Assignment: organization-defined authorities];	Functional	Intersects With	Contacts With Authorities	GOV-06	Mechanisms exist to identify and document appropriate contacts with relevant law enforcement and regulatory bodies	5	
IR-07	Incident Response Assistance	Provide an incident response support resource, integral to the organizational incident response capability, that offers advice and assistance to users of the system for the handling and reporting of incidents.	Functional	Equal	Incident Reporting Assistance	IRO-11	Mechanisms exist to provide incident response advice and assistance to users of Technology Assets, Applications and/or Devices (TAADs)	10	
IR-08	Incident Response Plan	a. Develop an incident response plan that:1. Provides the organization with a roadmap for implementing its incident response capability;2. Describes the structure and organization of the incident response capability;3. Provides a	Functional	Equal	Incident Response Plan (IRP)	IRO-04	Mechanisms exist to maintain and make available a current and viable Incident Response Plan (IRP) to all stakeholders	10	
MA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] maintenance policy that a. Addresses purpose	Functional	Subset Of	Maintenance Operations	MNT-01	Mechanisms exist to develop, disseminate, review & update procedures to facilitate the implementation of maintenance	10	
MA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] maintenance policy that a. Addresses purpose	Functional	Intersects With	Remote Maintenance Notifications	MNT-05.2	Mechanisms exist to require maintenance personnel to notify affected stakeholders when remote, non-local maintenance is planned	5	
MA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] maintenance policy that a. Addresses purpose	Functional	Intersects With	Auditing Remote Maintenance	MNT-05.1	Mechanisms exist to audit remote, non-local maintenance and diagnostic sessions, as well as review the maintenance action performed	5	
MA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] maintenance policy that a. Addresses purpose	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies,	5	
MA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] maintenance policy that a. Addresses purpose	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
MA-02	Controlled Maintenance	a. Schedule, document, and review records of maintenance, repair, and replacement on system components in accordance with manufacturer or vendor specifications and/or organizational requirements; b. Approve and monitor all	Functional	Equal	Controlled Maintenance	MNT-02	Mechanisms exist to conduct controlled maintenance activities throughout the lifecycle of the Technology Asset	10	
MA-04	Nonlocal Maintenance	a. Approve and monitor nonlocal maintenance and diagnostic activities; b. Allow the use of nonlocal maintenance and diagnostic tools only as consistent with organizational policy and documented in the security plan for the system; c. Employ	Functional	Intersects With	Remote Maintenance	MNT-05	Mechanisms exist to authorize, monitor and control remote, non-local maintenance and diagnostic activities	5	
MA-04	Nonlocal Maintenance	a. Approve and monitor nonlocal maintenance and diagnostic activities; b. Allow the use of nonlocal maintenance and diagnostic tools only as consistent with organizational policy and documented in the security plan for the system; c. Employ	Functional	Intersects With	Remote Maintenance Notifications	MNT-05.2	Mechanisms exist to require maintenance personnel to notify affected stakeholders when remote, non-local maintenance is planned	5	
MA-04	Nonlocal Maintenance	a. Approve and monitor nonlocal maintenance and diagnostic activities; b. Allow the use of nonlocal maintenance and diagnostic tools only as consistent with organizational policy and documented in the security plan for the system; c. Employ	Functional	Intersects With	Auditing Remote Maintenance	MNT-05.1	Mechanisms exist to audit remote, non-local maintenance and diagnostic sessions, as well as review the maintenance action performed	5	
MA-05	Maintenance Personnel	a. Establish a process for maintenance personnel authorization and maintain a list of authorized maintenance organizations or personnel; b. Verify that non-escorted personnel performing maintenance on the system possess the	Functional	Equal	Authorized Maintenance Personnel	MNT-06	Mechanisms exist to maintain a current list of authorized maintenance organizations or personnel	10	
MA-07	Field Maintenance	Restrict or prohibit field maintenance on [Assignment: organization-defined systems or system components] to [Assignment: organization-defined trusted maintenance facilities];	Functional	Equal	Field Maintenance	MNT-08	Mechanisms exist to securely conduct field maintenance on geographically deployed assets.	10	
MP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] media protection policy that a. Addresses	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures, at	5	
MP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] media protection policy that a. Addresses	Functional	Subset Of	Data Protection	DCH-01	Mechanisms exist to facilitate the implementation of data protection controls.	10	
MP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] media protection policy that a. Addresses	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
MP-02	Media Access	Restrict access to [Assignment: organization-defined types of digital and/or non-digital media] to [Assignment: organization-defined personnel or roles];	Functional	Intersects With	Media Access	DCH-03	Mechanisms exist to control and restrict access to digital and non-digital media to authorized individuals	5	
MP-02	Media Access	Restrict access to [Assignment: organization-defined types of digital and/or non-digital media] to [Assignment: organization-defined personnel or roles];	Functional	Intersects With	Endpoint Device Management (EDM)	END-01	Mechanisms exist to facilitate the implementation of Endpoint Device Management (EDM) controls.	5	
MP-06	Media Sanitization	a. Sanitize [Assignment: organization-defined system media] prior to disposal, release out of organizational control, or release for reuse using [Assignment: organization-defined sanitization techniques and procedures]; and b. Employ	Functional	Intersects With	Physical Media Disposal	DCH-08	Mechanisms exist to securely dispose of media when it is no longer required, using formal procedures.	5	
MP-06	Media Sanitization	a. Sanitize [Assignment: organization-defined system media] prior to disposal, release out of organizational control, or release for reuse using [Assignment: organization-defined sanitization techniques and procedures]; and b. Employ	Functional	Intersects With	System Media Sanitization	DCH-09	Mechanisms exist to sanitize system media with the strength and integrity commensurate with the classification or sensitivity of the information prior	5	

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
MP-06	Media Sanitization	a. Sanitize [Assignment: organization-defined system media] prior to disposal, release out of organizational control, or release for reuse using [Assignment: organization-defined sanitization techniques and procedures]; andb. Employ	Functional	Intersects With	Sanitization of Personal Data (PD)	DCH-09.3	Mechanisms exist to facilitate the sanitization of Personal Data (PD).	5	
MP-07	Media Use	a. [Selection (one): Restrict; Prohibit] the use of [Assignment: organization-defined types of system media] on [Assignment: organization-defined systems or system components] using [Assignment: organization-defined controls]; andb. Prohibit the	Functional	Intersects With	Media & Data Retention	DCH-18	Mechanisms exist to retain media and data in accordance with applicable statutory, regulatory and contractual obligations.	5	
MP-07	Media Use	a. [Selection (one): Restrict; Prohibit] the use of [Assignment: organization-defined types of system media] on [Assignment: organization-defined systems or system components] using [Assignment: organization-defined controls]; andb. Prohibit the	Functional	Intersects With	Media Use	DCH-10	Mechanisms exist to restrict the use of types of digital media on systems or system components.	5	
MP-07	Media Use	a. [Selection (one): Restrict; Prohibit] the use of [Assignment: organization-defined types of system media] on [Assignment: organization-defined systems or system components] using [Assignment: organization-defined controls]; andb. Prohibit the	Functional	Intersects With	Prohibit Use Without Owner	DCH-10.2	Mechanisms exist to prohibit the use of portable storage devices in organizational systems when such devices have identifiable owner	5	
PE-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] physical and environmental protection policy	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
PE-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] physical and environmental protection policy	Functional	Subset Of	Physical & Environmental Protections	PES-01	Mechanisms exist to facilitate the operation of physical and environmental protection controls.	10	
PE-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] physical and environmental protection policy	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies,	5	
PE-02	Physical Access Authorizations	a. Develop, approve, and maintain a list of individuals with authorized access to the facility where the system resides;b. Issue authorization credentials for facility access;c. Review the access list detailing authorized facility access by individuals	Functional	Equal	Physical Access Authorizations	PES-02	Physical access control mechanisms exist to maintain a current list of personnel with authorized access to organizational facilities (except for	10	
PE-03	Physical Access Control	a. Enforce physical access authorizations at [Assignment: organization-defined entry and exit points to the facility where the system resides] by;1. Verifying individual access authorizations before granting access to the facility; and2	Functional	Intersects With	Physical Access Control	PES-03	Physical access control mechanisms exist to enforce physical access authorizations for all physical access points (including designated	5	
PE-06	Monitoring Physical Access	a. Monitor physical access to the facility where the system resides to detect and respond to physical security incidents;b. Review physical access logs [Assignment: organization-defined frequency] and upon occurrence of [Assignment: organization-	Functional	Equal	Monitoring Physical Access	PES-05	Physical access control mechanisms exist to monitor for, detect and respond to physical security incidents	10	
PE-08	Visitor Access Records	a. Maintain visitor access records to the facility where the system resides for [Assignment: organization-defined time period];b. Review visitor access records [Assignment: organization-defined frequency]; andc. Report anomalies in	Functional	Equal	Physical Access Logs	PES-03.3	Physical access control mechanisms generate a log entry for each access attempt through controlled ingress and egress points	10	
PE-12	Emergency Lighting	Employ and maintain automatic emergency lighting for the system that activates in the event of a power outage or disruption and that covers emergency exits and evacuation routes within the facility	Functional	Equal	Emergency Lighting	PES-07.4	Facility security mechanisms exist to utilize and maintain automatic emergency lighting that activates in the event of a power outage or	10	
PE-13	Fire Protection	Employ and maintain fire detection and suppression systems that are supported by an independent energy source.	Functional	Equal	Fire Protection	PES-08	Facility security mechanisms exist to utilize and maintain fire suppression and detection devices/systems for the system that are supported by an	10	
PE-14	Environmental Controls	a. Maintain [Selection (one or more): temperature; humidity; pressure; radiation; [Assignment: organization-defined environmental control]] levels within the facility where the system resides at [Assignment: organization-defined	Functional	Equal	Temperature & Humidity Controls	PES-09	Facility security mechanisms exist to maintain and monitor temperature and humidity levels within the facility.	10	
PE-15	Water Damage Protection	Protect the system from damage resulting from water leakage by providing master shutoff or isolation valves that are accessible, working properly, and known to key personnel.	Functional	Equal	Water Damage Protection	PES-07.5	Facility security mechanisms exist to protect systems from damage resulting from water leakage by providing master shutoff valves that	10	
PE-16	Delivery and Removal	a. Authorize and control [Assignment: organization-defined types of system components] entering and exiting the facility; andb. Maintain records of the system components.	Functional	Equal	Delivery & Removal	PES-10	Physical security mechanisms exist to isolate information processing facilities from points such as delivery and loading areas and other points to	10	
PL-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] planning policy that;a. Addresses purpose	Functional	Subset Of	Statutory, Regulatory & Contractual Compliance	CPL-01	Mechanisms exist to facilitate the identification and implementation of relevant statutory, regulatory and contractual controls	10	
PL-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] planning policy that;a. Addresses purpose	Functional	Subset Of	Technology Development & Acquisition	TDA-01	Mechanisms exist to facilitate the implementation of tailored development and acquisition strategies, contract tools and	10	
PL-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] planning policy that;a. Addresses purpose	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies,	5	
PL-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] planning policy that;a. Addresses purpose	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
PL-02	System Security and Privacy Plans	a. Develop security and privacy plans for the system that;1. Are consistent with the organization's enterprise architecture;2. Explicitly define the constituent system components;3. Describe the operational context of the system in terms of	Functional	Intersects With	Plan / Coordinate with Other Organizational Entities	IAO-03.1	Mechanisms exist to plan and coordinate Information Assurance Program (IAP) activities with affected stakeholders before conducting such	5	
PL-02	System Security and Privacy Plans	a. Develop security and privacy plans for the system that;1. Are consistent with the organization's enterprise architecture;2. Explicitly define the constituent system components;3. Describe the operational context of the system in terms of	Functional	Intersects With	Applied Security, Compliance and Resilience Controls Documentation	IAO-03	Mechanisms exist to generate authoritative documentation (e.g., System Security Plan (SSP)) that;	5	
PL-02	System Security and Privacy Plans	a. Develop security and privacy plans for the system that;1. Are consistent with the organization's enterprise architecture;2. Explicitly define the constituent system components;3. Describe the operational context of the system in terms of	Functional	Intersects With	Network Diagrams & Data Flow Diagrams (DFDs)	AST-04	Mechanisms exist to maintain network architecture diagrams that: (1) Contain sufficient detail to assess the security of the network's	5	
PL-04	Rules of Behavior	a. Establish and provide to individuals requiring access to the system, the rules that describe their responsibilities and expected behavior for information and system usage, security, and privacy;b. Receive a documented acknowledgment from	Functional	Intersects With	Terms of Employment	HRS-05	Mechanisms exist to require all employees and contractors to apply cybersecurity and data protection principles in their daily work to enable	5	
PL-04	Rules of Behavior	a. Establish and provide to individuals requiring access to the system, the rules that describe their responsibilities and expected behavior for information and system usage, security, and privacy;b. Receive a documented acknowledgment from	Functional	Intersects With	Rules of Behavior	HRS-05.1	Mechanisms exist to define acceptable and unacceptable rules of behavior for the use of technologies, including consequences for	5	

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
PL-04	Rules of Behavior	a. Establish and provide to individuals requiring access to the system, the rules that describe their responsibilities and expected behavior for information and system usage, security, and privacy; b. Receive a documented acknowledgement from individuals that they understand and agree to the rules of behavior, restrictions on a. Use of social media, social networking sites, and external sites/applications; b. Posting organizational information on public websites; and c. Use of organization-provided identifiers.	Functional	Intersects With	Technology Use Restrictions	HRS-05.3	Mechanisms exist to establish usage restrictions and implementation guidance for organizational technologies based on the potential for misuse.	5	
PL-04(01)	Rules of Behavior Social Media and External Site/application Usage	Include in the rules of behavior, restrictions on a. Use of social media, social networking sites, and external sites/applications; b. Posting organizational information on public websites; and c. Use of organization-provided identifiers.	Functional	Equal	Social Media & Social Networking Restrictions	HRS-05.2	Mechanisms exist to define rules of behavior that contain explicit restrictions on the use of social media and networking sites, posting organizational information on public websites, and use of organization-provided identifiers.	10	
PL-10	Baseline Selection	Select a control baseline for the system.	Functional	Equal	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for systems.	10	
PL-11	Baseline Tailoring	Tailor the selected control baseline by applying specified tailoring actions.	Functional	Equal	Baseline Tailoring	CFG-02.9	Mechanisms exist to allow baseline controls to be specialized or customized by applying a defined set of tailoring actions that are specific to the system.	10	
PM-01	Information Security Program Plan	a. Develop and disseminate an organization-wide information security program plan that: 1. Provides an overview of the requirements for the security program and a description of the security program management controls and common controls.	Functional	Subset Of	Security, Compliance & Resilience Program (SCRPP)	GOV-01	Mechanisms exist to facilitate the implementation of security, compliance and resilience governance controls.	10	
PM-01	Information Security Program Plan	a. Develop and disseminate an organization-wide information security program plan that: 1. Provides an overview of the requirements for the security program and a description of the security program management controls and common controls.	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures at least annually.	5	
PM-01	Information Security Program Plan	a. Develop and disseminate an organization-wide information security program plan that: 1. Provides an overview of the requirements for the security program and a description of the security program management controls and common controls.	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient systems.	5	
PM-02	Information Security Program Leadership Role	Appoint a senior agency information security officer with the mission and resources to coordinate, develop, implement, and maintain an organization-wide information security program.	Functional	Intersects With	Assigned Security, Compliance & Resilience Responsibilities	GOV-04	Mechanisms exist to assign one or more qualified individuals with the mission and resources to centrally manage, coordinate, develop, implement, and maintain an organization-wide information security program.	5	
PM-03	Information Security and Privacy Resources	a. Include the resources needed to implement the information security and privacy programs in capital planning and investment requests and document all exceptions to this requirement; b. Prepare documentation required for addressing exceptions.	Functional	Equal	Security, Compliance & Resilience Resource Management	PRM-02	Mechanisms exist to address all capital planning and investment requests, including the resources needed to implement the Security, Compliance & Resilience Program.	10	
PM-04	Plan of Action and Milestones Process	a. Implement a process to ensure that plans of action and milestones for the information security, privacy, and supply chain risk management programs and associated organizational systems: 1. Are developed and maintained; 2. Address the mission and resources to coordinate, develop, implement, and maintain an organization-wide information security program.	Functional	Intersects With	Vulnerability Remediation Process	VPM-02	Mechanisms exist to ensure that vulnerabilities are properly identified, tracked and remediated.	5	
PM-04	Plan of Action and Milestones Process	a. Implement a process to ensure that plans of action and milestones for the information security, privacy, and supply chain risk management programs and associated organizational systems: 1. Are developed and maintained; 2. Address the mission and resources to coordinate, develop, implement, and maintain an organization-wide information security program.	Functional	Intersects With	Capabilities Deficiency Tracking	IAO-05	Mechanisms exist to govern identified deficiencies (e.g., Plan of Action and Milestones (POA&M) or similar methodology) that formally track and remediate deficiencies.	5	
PM-05	System Inventory	Develop and update [Assignment: organization-defined frequency] an inventory of organizational systems.	Functional	Intersects With	Asset Governance	AST-01	Mechanisms exist to facilitate an IT Asset Management (ITAM) program to implement and manage asset management controls.	5	
PM-05	System Inventory	Develop and update [Assignment: organization-defined frequency] an inventory of organizational systems.	Functional	Intersects With	Asset Inventories	AST-02	Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data (ITADSP) that are consistent with the organization's mission and resources.	5	
PM-06	Measures of Performance	Develop, monitor, and report on the results of information security and privacy measures of performance.	Functional	Intersects With	Assigned Security, Compliance & Resilience Responsibilities	GOV-04	Mechanisms exist to assign one or more qualified individuals with the mission and resources to centrally manage, coordinate, develop, implement, and maintain an organization-wide information security program.	5	
PM-06	Measures of Performance	Develop, monitor, and report on the results of information security and privacy measures of performance.	Functional	Intersects With	Measures of Performance	GOV-05	Mechanisms exist to develop, report and monitor Security, Compliance & Resilience Program (SCRPP) measures of performance.	5	
PM-07	Enterprise Architecture	Develop and maintain an enterprise architecture with consideration for information security, privacy, and the resulting risk to organizational operations and assets, individuals, other organizations, and the Nation.	Functional	Intersects With	Alignment With Enterprise Architecture	SEA-02	Mechanisms exist to develop an enterprise architecture, aligned with industry-recognized leading practices, with consideration for information security, privacy, and the resulting risk to organizational operations and assets, individuals, other organizations, and the Nation.	5	
PM-08	Critical Infrastructure Plan	Address information security and privacy issues in the development, documentation, and updating of a critical infrastructure and key resources protection plan.	Functional	Intersects With	Business Continuity Management System (BCMS)	BCD-01	Mechanisms exist to facilitate the implementation of contingency planning controls to help ensure resilient Technology Assets.	5	
PM-08	Critical Infrastructure Plan	Address information security and privacy issues in the development, documentation, and updating of a critical infrastructure and key resources protection plan.	Functional	Intersects With	Statutory, Regulatory & Contractual Compliance	CPL-01	Mechanisms exist to facilitate the identification and implementation of relevant statutory, regulatory and contractual controls.	5	
PM-09	Risk Management Strategy	a. Develops a comprehensive strategy to manage: 1. Security risk to organizational operations and assets, individuals, other organizations, and the Nation associated with the operation and use of organizational systems; and 2. Privacy risk to individuals, other organizations, and the Nation; and b. Manage the security and privacy state of organizational systems and the environments in which those systems operate through authorization processes; b. Designate individuals to fulfill specific roles and responsibilities within the organization.	Functional	Equal	Risk Management Program	RSK-01	Mechanisms exist to facilitate the implementation of strategic, operational and tactical risk management controls that are consistent with the organization's mission and resources.	10	
PM-10	Authorization Process	a. Manage the security and privacy state of organizational systems and the environments in which those systems operate through authorization processes; b. Designate individuals to fulfill specific roles and responsibilities within the organization.	Functional	Equal	Information Assurance (IA) Operations	IAO-01	Mechanisms exist to facilitate the implementation of security, compliance and resilience assessment and authorization controls.	10	
PM-11	Mission and Business Process Definition	a. Define organizational mission and business processes with consideration for information security and privacy and the resulting risk to organizational operations, organizational assets, individuals, other organizations, and the Nation; and b. Manage the security and privacy state of organizational systems and the environments in which those systems operate through authorization processes; b. Designate individuals to fulfill specific roles and responsibilities within the organization.	Functional	Equal	Business Process Definition	PRM-06	Mechanisms exist to define business processes with consideration for security, compliance and resilience that determines the organization's mission and resources.	10	
PM-12	Insider Threat Program	Implement an insider threat program that includes a cross-discipline insider threat incident handling team.	Functional	Equal	Insider Threat Program	THR-04	Mechanisms exist to implement an insider threat program that includes a cross-discipline insider threat incident handling team.	10	
PM-13	Security and Privacy Workforce	Establish a security and privacy workforce development and improvement program.	Functional	Intersects With	Defined Roles & Responsibilities	HRS-03	Mechanisms exist to define cybersecurity roles & responsibilities for all personnel.	5	
PM-13	Security and Privacy Workforce	Establish a security and privacy workforce development and improvement program.	Functional	Intersects With	Security, Compliance & Resilience-Minded Workforce	SAT-01	Mechanisms exist to facilitate the implementation of security workforce development and awareness controls.	5	
PM-14	Testing, Training, and Monitoring	a. Implement a process for ensuring that organizational plans for conducting security and privacy testing, training, and monitoring activities associated with organizational systems: 1. Are developed and maintained; and 2. Continue to be executed.	Functional	Intersects With	Personal Data (PD) Control Testing, Training & Monitoring	PRI-08	Mechanisms exist to conduct testing, training and monitoring activities for Personal Data (PD) controls.	5	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Over**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3-low.1>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
PM-14	Testing, Training, and Monitoring	a. Implement a process for ensuring that organizational plans for conducting security and privacy testing, training, and monitoring activities associated with organizational systems:1. Are developed and maintained; and 2. Continue to be executed;	Functional	Intersects With	Security, Compliance & Resilience Controls Oversight	CPL-02	Mechanisms exist to provide a security, compliance and resilience controls oversight function that reports to the organization's executive	5	
PM-15	Security and Privacy Groups and Associations	Establish and institutionalize contact with selected groups and associations within the security and privacy communities:a. To facilitate ongoing security and privacy education and training for organizational personnel;b. To maintain currency with	Functional	Intersects With	Threat Intelligence Program	THR-01	Mechanisms exist to implement a threat intelligence program that includes a cross-organization information-sharing capability that	5	
PM-15	Security and Privacy Groups and Associations	Establish and institutionalize contact with selected groups and associations within the security and privacy communities:a. To facilitate ongoing security and privacy education and training for organizational personnel;b. To maintain currency with	Functional	Intersects With	Contacts With Groups & Associations	GOV-07	Mechanisms exist to establish contact with selected groups and associations within the security, compliance and resilience	5	
PM-16	Threat Awareness Program	Implement a threat awareness program that includes a cross-organization information-sharing capability for threat intelligence.	Functional	Intersects With	Threat Intelligence Program	THR-01	Mechanisms exist to implement a threat intelligence program that includes a cross-organization information-sharing capability that	5	
PM-17	Protecting Controlled Unclassified Information on External Systems	a. Establish policy and procedures to ensure that requirements for the protection of controlled unclassified information that is processed, stored or transmitted on external systems, are implemented in accordance with applicable laws, executive	Functional	Equal	Protecting Sensitive / Regulated Data on External Technology Assets, Applications	DCH-13.3	Mechanisms exist to ensure that the requirements for the protection of sensitive and/or regulated data are maintained and transmitted as	10	
PM-18	Privacy Program Plan	a. Develop and disseminate an organization-wide privacy program plan that provides an overview of the agency's privacy program, and:1. Includes a description of the structure of the privacy program and the resources dedicated to the privacy	Functional	Equal	Data Privacy Program	PRI-01	Mechanisms exist to facilitate the implementation and operation of data protection controls throughout the data lifecycle to ensure all forms of	10	
PM-19	Privacy Program Leadership Role	Appoint a senior agency official for privacy with the authority, mission, accountability, and resources to coordinate, develop, and implement, applicable privacy requirements and manage privacy risks through the organization-wide privacy program	Functional	Equal	Chief Privacy Officer (CPO)	PRI-01.1	Mechanisms exist to appoints a Chief Privacy Officer (CPO) or similar role, with the authority, mission, accountability and resources to	10	
PM-20	Dissemination of Privacy Program Information	Maintain a central resource webpage on the organization's principal public website that serves as a central source of information about the organization's privacy program and that a. Ensures that the public has access to information about	Functional	Equal	Dissemination of Data Privacy Program Information	PRI-01.3	Mechanisms exist to: (1) Ensure that the public has access to information about organizational data privacy activities and can	10	
PM-20(01)	Dissemination of Privacy Program Information Privacy Policies on Websites	Develop and post privacy policies on all external-facing websites, mobile applications, and other digital services, that:a. Are written in plain language and organized in a way that is easy to understand and navigate;b. Provide information	Functional	Equal	Data Privacy Notice	PRI-02	Mechanisms exist to: (1) Make data privacy notice(s) available to individuals upon first interaction with an organization and	10	
PM-21	Accounting of Disclosures	a. Develop and maintain an accurate accounting of disclosures of personally identifiable information, including:1. Date, nature, and purpose of each disclosure; and2. Name and address, or other contact information of the individual or	Functional	Equal	Accounting of Disclosures	PRI-14.1	Mechanisms exist to provide data subjects with an accounting of disclosures of their Personal Data (PD) controlled by	10	
PM-22	Personally Identifiable Information Quality Management	Develop and document organization-wide policies and procedures for:a. Reviewing for the accuracy, relevance, timeliness, and completeness of personally identifiable information across the information life cycle;b. Correcting or	Functional	Intersects With	Data Quality Management	PRI-10	Mechanisms exist to manage the quality, utility, objectivity, integrity and impact determination and de- identification of sensitive and/or	5	
PM-22	Personally Identifiable Information Quality Management	Develop and document organization-wide policies and procedures for:a. Reviewing for the accuracy, relevance, timeliness, and completeness of personally identifiable information across the information life cycle;b. Correcting or	Functional	Intersects With	Data Quality Operations	DCH-22	Mechanisms exist to check for Redundant, Obsolete/Outdated, Toxic or Trivial (ROTT) data to ensure the accuracy, relevance, timeliness	5	
PM-23	Data Governance Body	Establish a Data Governance Body consisting of [Assignment: organization-defined roles] with [Assignment: organization-defined responsibilities].	Functional	Intersects With	Data Management Board	PRI-13	Mechanisms exist to establish a written charter for a Data Management Board (DMB) and assigned organization-defined roles	5	
PM-23	Data Governance Body	Establish a Data Governance Body consisting of [Assignment: organization-defined roles] with [Assignment: organization-defined responsibilities].	Functional	Intersects With	Data Quality Management	PRI-10	Mechanisms exist to manage the quality, utility, objectivity, integrity and impact determination and de- identification of sensitive and/or	5	
PM-23	Data Governance Body	Establish a Data Governance Body consisting of [Assignment: organization-defined roles] with [Assignment: organization-defined responsibilities].	Functional	Intersects With	Data Governance	GOV-10	Mechanisms exist to facilitate data governance to oversee the organization's policies, standards and procedures so that sensitive and/or	5	
PM-24	Data Integrity Board	Establish a Data Integrity Board to:a. Review proposals to conduct or participate in a matching program; andb. Conduct an annual review of all matching programs in which the agency has participated	Functional	Intersects With	Data Governance	GOV-10	Mechanisms exist to facilitate data governance to oversee the organization's policies, standards and procedures so that sensitive and/or	5	
PM-24	Data Integrity Board	Establish a Data Integrity Board to:a. Review proposals to conduct or participate in a matching program; andb. Conduct an annual review of all matching programs in which the agency has participated	Functional	Intersects With	Data Management Board	PRI-13	Mechanisms exist to establish a written charter for a Data Management Board (DMB) and assigned organization-defined roles	5	
PM-24	Data Integrity Board	Establish a Data Integrity Board to:a. Review proposals to conduct or participate in a matching program; andb. Conduct an annual review of all matching programs in which the agency has participated	Functional	Intersects With	Data Quality Management	PRI-10	Mechanisms exist to manage the quality, utility, objectivity, integrity and impact determination and de- identification of sensitive and/or	5	
PM-24	Data Integrity Board	Establish a Data Integrity Board to:a. Review proposals to conduct or participate in a matching program; andb. Conduct an annual review of all matching programs in which the agency has participated	Functional	Intersects With	Personal Data (PD) Accuracy & Integrity	PRI-05.2	Mechanisms exist to ensure the accuracy and relevance of Personal Data (PD) throughout the information lifecycle	5	
PM-24	Data Integrity Board	Establish a Data Integrity Board to:a. Review proposals to conduct or participate in a matching program; andb. Conduct an annual review of all matching programs in which the agency has participated	Functional	Intersects With	Computer Matching Agreements (CMA)	PRI-02.3	Mechanisms exist to publish Computer Matching Agreements (CMA) on the organization's public website(s)	5	
PM-24	Data Integrity Board	Establish a Data Integrity Board to:a. Review proposals to conduct or participate in a matching program; andb. Conduct an annual review of all matching programs in which the agency has participated	Functional	Intersects With	Automated Data Management Processes	PRI-02.2	Automated mechanisms exist to adjust data that is able to be collected, received, processed, stored, transmitted, shared, updated	5	
PM-25	Minimization of Personally Identifiable Information Used in Testing, Training, and	a. Develop, document, and implement policies and procedures that address the use of personally identifiable information for internal testing, training, and research;b. Limit or minimize the amount of personally identifiable information used for internal	Functional	Intersects With	Usage Restrictions of Personal Data (PD)	PRI-05.4	Mechanisms exist to restrict collecting, receiving, processing, storing, transmitting, sharing and/or modifying Personal Data (PD)	5	
PM-25	Minimization of Personally Identifiable Information Used in Testing, Training, and	a. Develop, document, and implement policies and procedures that address the use of personally identifiable information for internal testing, training, and research;b. Limit or minimize the amount of personally identifiable information used for internal	Functional	Intersects With	Collection Minimization	END-13.3	Mechanisms exist to utilize sensors that are configured to minimize the collection of information about individuals	5	
PM-25	Minimization of Personally Identifiable Information Used in Testing, Training, and	a. Develop, document, and implement policies and procedures that address the use of personally identifiable information for internal testing, training, and research;b. Limit or minimize the amount of personally identifiable information used for internal	Functional	Intersects With	Minimize Visitor Personal Data (PD)	PES-06.5	Mechanisms exist to minimize the collection of Personal Data (PD) contained in visitor access records.	5	
PM-25	Minimization of Personally Identifiable Information Used in Testing, Training, and	a. Develop, document, and implement policies and procedures that address the use of personally identifiable information for internal testing, training, and research;b. Limit or minimize the amount of personally identifiable information used for internal	Functional	Intersects With	Internal Use of Personal Data (PD) For Testing, Training and Research	PRI-05.1	Mechanisms exist to address the use of Personal Data (PD) for internal testing, training and research that: (1) Takes measures to limit ex	5	

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Focal Document: **NIST SP 800 - R2 R3 - Guide to Operational Technology (OT) Security - Low OT Over**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

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FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
PM-25	Minimization of Personally Identifiable Information Used in Testing, Training, and	a. Develop, document, and implement policies and procedures that address the use of personally identifiable information for internal testing, training, and research;b. Limit or minimize the amount of personally identifiable information used for internal	Functional	Intersects With	Limit Sensitive / Regulated Data In Testing, Training & Research	DCH-18.2	Mechanisms exist to minimize the use of sensitive and/or regulated data for research, testing, or training, in accordance with authorized	5	
PM-26	Complaint Management	Implement a process for receiving and responding to complaints, concerns, or questions from individuals about the organizational security and privacy practices that includes:a. Mechanisms that are easy to use and readily accessible by the	Functional	Intersects With	Data Subject Feedback Management	PRI-06.4	Mechanisms exist to maintain a process to efficiently and effectively respond to requests, complaints, concerns and/or questions from	5	
PM-26	Complaint Management	Implement a process for receiving and responding to complaints, concerns, or questions from individuals about the organizational security and privacy practices that includes:a. Mechanisms that are easy to use and readily accessible by the	Functional	Intersects With	Appeal Adverse Decision	PRI-06.3	Mechanisms exist to maintain a process for data subjects to appeal an adverse decision.	5	
PM-27	Privacy Reporting	a. Develop [Assignment: organization-defined privacy reports] and disseminate to:1. [Assignment: organization-defined oversight bodies] to demonstrate accountability with statutory, regulatory, and policy privacy mandates; and2. [Assignment:	Functional	Equal	Documenting Data Processing Activities	PRI-14	Mechanisms exist to document Personal Data (PD) processing activities that covers collection,	10	
PM-28	Risk Framing	a. Identify and document:1. Assumptions affecting risk assessments, risk responses, and risk monitoring;2. Constraints affecting risk assessments, risk responses, and risk monitoring;3. Priorities and trade-offs considered by the	Functional	Equal	Risk Framing	RSK-01.1	Mechanisms exist to identify: (1) Assumptions affecting risk assessments, risk response and risk monitoring;	10	
PM-29	Risk Management Program Leadership Roles	a. Appoint a Senior Accountable Official for Risk Management to align organizational information security and privacy management processes with strategic, operational, and budgetary planning processes; andb. Establish a Risk Executive	Functional	Intersects With	Supply Chain Risk Management (SCRM) Plan	RSK-09	Mechanisms exist to develop a plan for Supply Chain Risk Management (SCRM) associated with the development activities	5	
PM-29	Risk Management Program Leadership Roles	a. Appoint a Senior Accountable Official for Risk Management to align organizational information security and privacy management processes with strategic, operational, and budgetary planning processes; andb. Establish a Risk Executive	Functional	Intersects With	Assigned Security, Compliance & Resilience Responsibilities	GOV-04	Mechanisms exist to assign one or more qualified individuals with the mission and resources to centrally-manage, coordinate, develop	5	
PM-29	Risk Management Program Leadership Roles	a. Appoint a Senior Accountable Official for Risk Management to align organizational information security and privacy management processes with strategic, operational, and budgetary planning processes; andb. Establish a Risk Executive	Functional	Intersects With	Risk Management Program	RSK-01	Mechanisms exist to facilitate the implementation of strategic, operational and tactical risk management controls that are	5	
PM-30	Supply Chain Risk Management Strategy	a. Develop an organization-wide strategy for managing supply chain risks associated with the development, acquisition, maintenance, and disposal of systems, system components, and system services;b. Implement the supply chain risk	Functional	Equal	Supply Chain Risk Management (SCRM) Plan	RSK-09	Mechanisms exist to develop a plan for Supply Chain Risk Management (SCRM) associated with the development activities	10	
PM-30(01)	Supply Chain Risk Management Strategy Suppliers of Critical or Mission-essential	Identify, prioritize, and assess suppliers of critical or mission-essential technologies, products, and services.	Functional	Intersects With	Customized Development of Critical Components	TDA-12	Mechanisms exist to custom-develop critical system components, when Commercial Off The Shelf (COTS) solutions are unavailable.	5	
PM-30(01)	Supply Chain Risk Management Strategy Suppliers of Critical or Mission-essential	Identify, prioritize, and assess suppliers of critical or mission-essential technologies, products, and services.	Functional	Intersects With	Criticality Analysis During Development	TDA-06.1	Mechanisms exist to require the developer of the Technology Asset, Application and/or Service (TAAS) to	5	
PM-30(01)	Supply Chain Risk Management Strategy Suppliers of Critical or Mission-essential	Identify, prioritize, and assess suppliers of critical or mission-essential technologies, products, and services.	Functional	Intersects With	Third-Party Criticality Assessments	TPM-02	Mechanisms exist to identify, prioritize and assess suppliers and partners of critical Technology Assets, Applications and/or Services	5	
PM-31	Continuous Monitoring Strategy	Develop an organization-wide continuous monitoring strategy and implement continuous monitoring programs that include:a. Establishing the following organization-wide metrics to be monitored: [Assignment: organization-defined metrics];b. Analyze [Assignment: organization-defined systems or systems components] supporting mission essential services or functions to ensure that the information resources are being used consistent with their intended purpose.	Functional	Subset Of	Continuous Monitoring	MON-01	Mechanisms exist to facilitate the implementation of enterprise-wide monitoring controls.	10	
PM-32	Purposing	Analyze [Assignment: organization-defined systems or systems components] supporting mission essential services or functions to ensure that the information resources are being used consistent with their intended purpose.	Functional	Equal	Purpose Validation	GOV-11	Mechanisms exist to monitor mission/business-critical Technology Assets, Applications and/or Services (TAAS) to ensure those resources are	10	
PS-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] personnel security policy that a. Addresses	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
PS-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] personnel security policy that a. Addresses	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCR), including policies, standards and procedures, at	5	
PS-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] personnel security policy that a. Addresses	Functional	Subset Of	Human Resources Security Management	HRS-01	Mechanisms exist to facilitate the implementation of personnel security controls.	10	
PS-02	Position Risk Designation	a. Assign a risk designation to all organizational positions;b. Establish screening criteria for individuals filling those positions; andc. Review and update position risk designations [Assignment: organization-defined frequency].	Functional	Intersects With	Competency Requirements for Security-Related Positions	HRS-03.2	Mechanisms exist to ensure that all security-related positions are staffed by qualified individuals who have the necessary skill set	5	
PS-02	Position Risk Designation	a. Assign a risk designation to all organizational positions;b. Establish screening criteria for individuals filling those positions; andc. Review and update position risk designations [Assignment: organization-defined frequency].	Functional	Intersects With	Position Categorization	HRS-02	Mechanisms exist to manage personnel security risk by assigning a risk designation to all positions and establishing screening criteria for	5	
PS-03	Personnel Screening	a. Screen individuals prior to authorizing access to the system; andb. Rescreen individuals in accordance with [Assignment: organization-defined conditions requiring rescreening and, where rescreening is so indicated, the frequency of	Functional	Equal	Personnel Screening	HRS-04	Mechanisms exist to manage personnel security risk by screening individuals prior to authorizing access	10	
PS-04	Personnel Termination	Upon termination of individual employment:a. Disable system access within [Assignment: organization-defined time period];b. Terminate or revoke any authenticators and credentials associated with the individual;c. Conduct exit	Functional	Equal	Personnel Termination	HRS-09	Mechanisms exist to govern the termination of individual employment.	10	
PS-05	Personnel Transfer	a. Review and confirm ongoing operational need for current logical and physical access authorizations to systems and facilities when individuals are reassigned or transferred to other positions within the organization;b. Initiate [Assignment:	Functional	Equal	Personnel Transfer	HRS-08	Mechanisms exist to adjust logical and physical access authorizations to Technology Assets, Applications and/or Services (TAAS) and facilities	10	
PS-06	Access Agreements	a. Develop and document access agreements for organizational systems;b. Review and update the access agreements [Assignment: organization-defined frequency]; andc. Verify that individuals requiring access to organizational	Functional	Intersects With	Confidentiality Agreements	HRS-06.1	Mechanisms exist to require Non-Disclosure Agreements (NDAs) or similar confidentiality agreements that reflect the needs to protect data	5	
PS-06	Access Agreements	a. Develop and document access agreements for organizational systems;b. Review and update the access agreements [Assignment: organization-defined frequency]; andc. Verify that individuals requiring access to organizational	Functional	Intersects With	Access Agreements	HRS-06	Mechanisms exist to require internal and third-party users to sign appropriate access agreements prior to having granted access	5	
PS-07	External Personnel Security	a. Establish personnel security requirements, including security roles and responsibilities for external providers;b. Require external providers to comply with personnel security policies and procedures established by the organization;c.	Functional	Equal	Third-Party Personnel	HRS-10	Mechanisms exist to govern third-party personnel by reviewing and monitoring third-party security, compliance and/or resilience roles	10	

Secure Controls Framework (SCF)

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Over**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3-low_1

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
SA-04(10)	Acquisition Process Use of Approved PIV Products	Employ only information technology products on the FIPS 201-approved products list for Personal Identity Verification (PIV) capability implemented within organizational systems.	Functional	Intersects With	Information Assurance Enabled Products	TDA-02.2	Mechanisms exist to limit the use of commercially-provided Information Assurance (IA) and IA-enabled IT products to those products that have	5	
SA-04(12)	Acquisition Process Data Ownership	a. Include organizational data ownership requirements in the acquisition contract; andb. Require all data to be removed from the contractor's system and returned to the organization within [Assignment: organization-defined time frame].	Functional	Intersects With	Personal Data (PD) Lineage	PRI-09	Mechanisms exist to maintain a process to document the lineage of Personal Data (PD) by recording how the organization collects, receives,	5	
SA-04(12)	Acquisition Process Data Ownership	a. Include organizational data ownership requirements in the acquisition contract; andb. Require all data to be removed from the contractor's system and returned to the organization within [Assignment: organization-defined time frame].	Functional	Intersects With	Data Stewardship	DCH-01.1	Mechanisms exist to ensure data stewardship is assigned, documented and communicated.	5	
SA-04(12)	Acquisition Process Data Ownership	a. Include organizational data ownership requirements in the acquisition contract; andb. Require all data to be removed from the contractor's system and returned to the organization within [Assignment: organization-defined time frame].	Functional	Intersects With	Asset Ownership Assignment	AST-03	Mechanisms exist to ensure asset ownership responsibilities are assigned, tracked and managed at a team, individual, or responsible	5	
SA-05	System Documentation	a. Obtain or develop administrator documentation for the system, system component, or system service that describes:1. Secure configuration, installation, and operation of the system, component, or service;2. Effective use and	Functional	Intersects With	Documentation Requirements	TDA-04	Mechanisms exist to obtain, protect and distribute administrator documentation for Technology Assets, Applications and/or Services	5	
SA-05	System Documentation	a. Obtain or develop administrator documentation for the system, system component, or system service that describes:1. Secure configuration, installation, and operation of the system, component, or service;2. Effective use and	Functional	Intersects With	Asset Scope Classification	AST-04.1	Mechanisms exist to determine security, compliance and resilience control applicability by identifying, assessing and documenting the	5	
SA-08	Security and Privacy Engineering Principles	Apply the following systems security and privacy engineering principles in the specification, design, development, implementation, and modification of the system and system components: [Assignment: organization-defined systems	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for	5	
SA-08	Security and Privacy Engineering Principles	Apply the following systems security and privacy engineering principles in the specification, design, development, implementation, and modification of the system and system components: [Assignment: organization-defined systems	Functional	Intersects With	Secure Engineering Principles	SEA-01	Mechanisms exist to facilitate the implementation of industry-recognized security, compliance and	5	
SA-09	External System Services	a. Require that providers of external system services comply with organizational security and privacy requirements and employ the following controls: [Assignment: organization-defined controls];b. Define and document organizational	Functional	Equal	Third-Party Services	TPM-04	Mechanisms exist to mitigate the risks associated with third-party access to the organization's Technology Assets, Applications	10	
SA-22	Unsupported System Components	a. Replace system components when support for the components is no longer available from the developer, vendor, or manufacturer; orb. Provide the following options for alternative sources for continued support for unsupported	Functional	Intersects With	Unsupported Technology Assets, Applications and/or Services (TAAS)	TDA-17	Mechanisms exist to prevent unsupported Technology Assets, Applications and/or Services (TAAS)	5	
SA-22	Unsupported System Components	a. Replace system components when support for the components is no longer available from the developer, vendor, or manufacturer; orb. Provide the following options for alternative sources for continued support for unsupported	Functional	Intersects With	Alternate Sources for Continued Support	TDA-17.1	Mechanisms exist to provide in-house support or contract external providers for support with unsupported	5	
SC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] system and communications protection policy	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
SC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] system and communications protection policy	Functional	Subset Of	Network Security Controls (NSC)	NET-01	Mechanisms exist to develop, govern & update procedures to facilitate the implementation of Network Security Controls (NSC)	10	
SC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] system and communications protection policy	Functional	Subset Of	Secure Engineering Principles	SEA-01	Mechanisms exist to facilitate the implementation of industry-recognized security, compliance and	10	
SC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] system and communications protection policy	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies,	5	
SC-05	Denial-of-service Protection	a. [Selection (one): Protect against; Limit] the effects of the following types of denial-of-service events: [Assignment: organization-defined types of denial-of-service events]; andb. Employ the following controls to achieve the denial-of-service	Functional	Intersects With	Resource Priority	CAP-02	Mechanisms exist to control resource utilization of Technology Assets, Applications and/or Services (TAAS) that are susceptible to Denial of	5	
SC-05	Denial-of-service Protection	a. [Selection (one): Protect against; Limit] the effects of the following types of denial-of-service events: [Assignment: organization-defined types of denial-of-service events]; andb. Employ the following controls to achieve the denial-of-service	Functional	Intersects With	Capacity Planning	CAP-03	Mechanisms exist to conduct capacity planning so that necessary capacity for information processing, telecommunications and	5	
SC-05	Denial-of-service Protection	a. [Selection (one): Protect against; Limit] the effects of the following types of denial-of-service events: [Assignment: organization-defined types of denial-of-service events]; andb. Employ the following controls to achieve the denial-of-service	Functional	Intersects With	Capacity & Performance Management	CAP-01	Mechanisms exist to facilitate the implementation of capacity management controls to ensure optimal system performance to meet	5	
SC-05	Denial-of-service Protection	a. [Selection (one): Protect against; Limit] the effects of the following types of denial-of-service events: [Assignment: organization-defined types of denial-of-service events]; andb. Employ the following controls to achieve the denial-of-service	Functional	Intersects With	Denial of Service (DoS) Protection	NET-02.1	Automated mechanisms exist to protect against or limit the effects of denial of service attacks.	5	
SC-07	Boundary Protection	a. Monitor and control communications at the external managed interfaces to the system and at key internal managed interfaces within the system;b. Implement subnetworks for publicly accessible system components that are [Selection	Functional	Intersects With	Boundary Protection	NET-03	Mechanisms exist to monitor and control communications at the external network boundary and at key internal boundaries within the	5	
SC-07(28)	Boundary Protection Connections to Public Networks	Prohibit the direct connection of [Assignment: organization-defined system] to a public network.	Functional	Equal	Direct Internet Access Restrictions	NET-06.5	Mechanisms exist to prohibit, or strictly-control, Internet access from sensitive and/or regulated data enclaves (secure zones).	10	
SC-07(29)	Boundary Protection Separate Subnets to Isolate Functions	Implement [Selection (one): physically; logically] separate subnetworks to isolate the following critical system components and functions: [Assignment: organization-defined critical system components and functions].	Functional	Intersects With	Cloud Infrastructure Security Subnet	CLD-03	Mechanisms exist to host security-specific technologies in a dedicated subnet.	5	
SC-12	Cryptographic Key Establishment and Management	Establish and manage cryptographic keys when cryptography is employed within the system in accordance with the following key management requirements: [Assignment: organization-defined requirements for key generation, distribution, storage,	Functional	Intersects With	Public Key Infrastructure (PKI)	CRY-08	Mechanisms exist to securely implement an internal Public Key Infrastructure (PKI) infrastructure or obtain PKI services from a reputable	5	
SC-13	Cryptographic Protection	a. Determine the [Assignment: organization-defined cryptographic uses]; andb. Implement the following types of cryptography required for each specified cryptographic use: [Assignment: organization-defined types of cryptography for	Functional	Intersects With	Encrypting Data At Rest	CRY-05	Cryptographic mechanisms exist to prevent unauthorized disclosure of data at rest.	5	
SC-13	Cryptographic Protection	a. Determine the [Assignment: organization-defined cryptographic uses]; andb. Implement the following types of cryptography required for each specified cryptographic use: [Assignment: organization-defined types of cryptography for	Functional	Intersects With	Export-Controlled Cryptography	CRY-01.2	Mechanisms exist to address the exporting of cryptographic technologies in compliance with relevant statutory and regulatory	5	

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
SC-13	Cryptographic Protection	a. Determine the [Assignment: organization-defined cryptographic uses]; andb. Implement the following types of cryptography required for each specified cryptographic use: [Assignment: organization-defined types of cryptography for a. Prohibit remote activation of collaborative computing devices and applications with the following exceptions: [Assignment: organization-defined exceptions where remote activation is to be allowed]; andb. Provide an explicit indication a. Provide additional data origin authentication and integrity verification artifacts along with the authoritative name resolution data the system returns in response to external name/address resolution queries; andb. Provide the means to	Functional	Intersects With	Use of Cryptographic Controls	CRY-01	Mechanisms exist to facilitate the implementation of cryptographic protections controls using known public standards and trusted	5	
SC-15	Collaborative Computing Devices and Applications	a. Prohibit remote activation of collaborative computing devices and applications with the following exceptions: [Assignment: organization-defined exceptions where remote activation is to be allowed]; andb. Provide an explicit indication a. Provide additional data origin authentication and integrity verification artifacts along with the authoritative name resolution data the system returns in response to external name/address resolution queries; andb. Provide the means to	Functional	Intersects With	Collaborative Computing Devices	END-14	Mechanisms exist to unplug or prohibit the remote activation of collaborative computing devices with the following exceptions:	5	
SC-20	Secure Name/address Resolution Service (authoritative Source)	a. Provide additional data origin authentication and integrity verification artifacts along with the authoritative name resolution data the system returns in response to external name/address resolution queries; andb. Provide the means to	Functional	Intersects With	Domain Name Service (DNS) Resolution	NET-10	Mechanisms exist to ensure Domain Name Service (DNS) resolution is designed, implemented and managed to protect the security of name /	5	
SC-21	Secure Name/address Resolution Service (recursive or Caching Resolver)	Request and perform data origin authentication and data integrity verification on the name/address resolution responses the system receives from authoritative sources.	Functional	Equal	Secure Name / Address Resolution Service (Recursive or Caching Resolver)	NET-10.2	Mechanisms exist to perform data origin authentication and data integrity verification on the Domain Name Service (DNS) resolution	10	
SC-22	Architecture and Provisioning for Name/address Resolution Service	Ensure the systems that collectively provide name/address resolution service for an organization are fault-tolerant and implement internal and external role separation.	Functional	Equal	Architecture & Provisioning for Name / Address Resolution Service	NET-10.1	Mechanisms exist to ensure systems that collectively provide Domain Name Service (DNS) resolution service are fault-tolerant and	10	
SC-39	Process Isolation	Maintain a separate execution domain for each executing system process.	Functional	Equal	Process Isolation	SEA-04	Mechanisms exist to implement a separate execution domain for each executing process.	10	
SC-41	Port and I/O Device Access	[Selection (one): Physically; Logically] disable or remove [Assignment: organization-defined connection ports or input/output devices] on the following systems or system components: [Assignment: organization-defined systems or	Functional	Equal	Port & Input / Output (I/O) Device Access	END-12	Mechanisms exist to physically disable or remove unnecessary connection ports or input/output devices from sensitive systems	10	
SC-45	System Time Synchronization	Synchronize system clocks within and between systems and system components.	Functional	Intersects With	Synchronization With Authoritative Time Source	MON-07.1	Mechanisms exist to synchronize internal system clocks with an authoritative time source.	5	
SI-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] system and information integrity policy that:a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] system and information integrity policy that:a	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures, at	5	
SI-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] system and information integrity policy that:a	Functional	Subset Of	Secure Engineering Principles	SEA-01	Mechanisms exist to facilitate the implementation of industry-recognized security, compliance and	10	
SI-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] system and information integrity policy that:a	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
SI-02	Flaw Remediation	a. Identify, report, and correct system flaws;b. Test software and firmware updates related to flaw remediation for effectiveness and potential side effects before installation;c. Install security-relevant software and firmware updates within	Functional	Intersects With	Vulnerability & Patch Management Program (VPM)	VPM-01	Mechanisms exist to facilitate the implementation and monitoring of vulnerability management controls.	5	
SI-02	Flaw Remediation	a. Identify, report, and correct system flaws;b. Test software and firmware updates related to flaw remediation for effectiveness and potential side effects before installation;c. Install security-relevant software and firmware updates within	Functional	Intersects With	Software & Firmware Patching	VPM-05	Mechanisms exist to conduct software patching for all deployed Technology Assets, Applications	5	
SI-02	Flaw Remediation	a. Identify, report, and correct system flaws;b. Test software and firmware updates related to flaw remediation for effectiveness and potential side effects before installation;c. Install security-relevant software and firmware updates within	Functional	Intersects With	Automatic Antimalware Signature Updates	END-04.1	Automated mechanisms exist to update antimalware technologies, including signature definitions.	5	
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code;b. Automatically update malicious code protection	Functional	Intersects With	Software & Firmware Patching	VPM-05	Mechanisms exist to conduct software patching for all deployed Technology Assets, Applications	5	
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code;b. Automatically update malicious code protection	Functional	Intersects With	Vulnerability & Patch Management Program (VPM)	VPM-01	Mechanisms exist to facilitate the implementation and monitoring of vulnerability management controls.	5	
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code;b. Automatically update malicious code protection	Functional	Intersects With	Malicious Code Protection (Anti-Malware)	END-04	Mechanisms exist to utilize antimalware technologies to detect and eradicate malicious code.	5	
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code;b. Automatically update malicious code protection	Functional	Intersects With	Heuristic / Nonsignature-Based Detection	END-04.4	Mechanisms exist to utilize heuristic / nonsignature-based antimalware detection capabilities.	5	
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code;b. Automatically update malicious code protection	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during	5	
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code;b. Automatically update malicious code protection	Functional	Intersects With	Automatic Antimalware Signature Updates	END-04.1	Automated mechanisms exist to update antimalware technologies, including signature definitions.	5	
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code;b. Automatically update malicious code protection	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5	
SI-04	System Monitoring	a. Monitor the system to detect:1. Attacks and indicators of potential attacks in accordance with the following monitoring objectives: [Assignment: organization-defined monitoring objectives]; and2. Unauthorized local network and remote	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5	
SI-04	System Monitoring	a. Monitor the system to detect:1. Attacks and indicators of potential attacks in accordance with the following monitoring objectives: [Assignment: organization-defined monitoring objectives]; and2. Unauthorized local network and remote	Functional	Intersects With	Centralized Collection of Security Event Logs	MON-02	Mechanisms exist to utilize a Security Incident Event Manager (SIEM), or similar automated tool, to support the centralized collection of security	5	
SI-04	System Monitoring	a. Monitor the system to detect:1. Attacks and indicators of potential attacks in accordance with the following monitoring objectives: [Assignment: organization-defined monitoring objectives]; and2. Unauthorized local network and remote	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during	5	
SI-04	System Monitoring	a. Monitor the system to detect:1. Attacks and indicators of potential attacks in accordance with the following monitoring objectives: [Assignment: organization-defined monitoring objectives]; and2. Unauthorized local network and remote	Functional	Intersects With	Continuous Monitoring	MON-01	Mechanisms exist to facilitate the implementation of enterprise-wide monitoring controls.	5	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Over**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3-low.1>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
SI-05	Security Alerts, Advisories, and Directives	a. Receive system security alerts, advisories, and directives from [Assignment: organization-defined external organizations] on an ongoing basis;b. Generate internal security alerts, advisories, and directives as deemed necessary;c.	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5	
SI-05	Security Alerts, Advisories, and Directives	a. Receive system security alerts, advisories, and directives from [Assignment: organization-defined external organizations] on an ongoing basis;b. Generate internal security alerts, advisories, and directives as deemed necessary;c.	Functional	Intersects With	Threat Intelligence Feeds	THR-03	Mechanisms exist to maintain situational awareness of vulnerabilities and evolving threats by leveraging the knowledge of attacker.	5	
SI-05	Security Alerts, Advisories, and Directives	a. Receive system security alerts, advisories, and directives from [Assignment: organization-defined external organizations] on an ongoing basis;b. Generate internal security alerts, advisories, and directives as deemed necessary;c.	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during transmission.	5	
SI-12	Information Management and Retention	Manage and retain information within the system and information output from the system in accordance with applicable laws, executive orders, directives, regulations, policies, standards, guidelines and operational requirements.	Functional	Intersects With	Personal Data (PD) Retention & Disposal	PRI-05	Mechanisms exist to: (1) Retain Personal Data (PD), including metadata, for an organization-defined time period to support business operations.	5	
SI-17	Fail-safe Procedures	Implement the indicated fail-safe procedures when the indicated failures occur: [Assignment: organization-defined list of failure conditions and associated fail-safe procedures].	Functional	Equal	Fail Safe	SEA-07.3	Mechanisms exist to implement fail-safe procedures when failure conditions occur.	10	
SR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] supply chain risk management policy that:a.	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures necessary for secure, compliant and resilient.	5	
SR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] supply chain risk management policy that:a.	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient.	5	
SR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] supply chain risk management policy that:a.	Functional	Subset Of	Third-Party Management	TPM-01	Mechanisms exist to facilitate the implementation of third-party management controls.	10	
SR-02	Supply Chain Risk Management Plan	a. Develop a plan for managing supply chain risks associated with the research and development, design, manufacturing, acquisition, delivery, integration, operations and maintenance, and disposal of the following systems, system components or	Functional	Intersects With	Supply Chain Risk Management (SCRM) Plan	RSK-09	Mechanisms exist to develop a plan for Supply Chain Risk Management (SCRM) associated with the development, acquisition,	5	
SR-02	Supply Chain Risk Management Plan	a. Develop a plan for managing supply chain risks associated with the research and development, design, manufacturing, acquisition, delivery, integration, operations and maintenance, and disposal of the following systems, system components or	Functional	Intersects With	Supply Chain Risk Management (SCRM)	TPM-03	Mechanisms exist to: (1) Evaluate security risks and threats associated with Technology Assets, Acquisition and/or Services (TAAS).	5	
SR-02(01)	Supply Chain Risk Management Plan Establish SCRM Team	Establish a supply chain risk management team consisting of [Assignment: organization-defined personnel, roles, and responsibilities] to lead and support the following SCRM activities: [Assignment: organization-defined supply chain risk	Functional	Intersects With	Supply Chain Risk Management (SCRM)	TPM-03	Mechanisms exist to: (1) Evaluate security risks and threats associated with Technology Assets, Acquisition and/or Services (TAAS).	5	
SR-03	Supply Chain Controls and Processes	a. Establish a process or processes to identify and address weaknesses or deficiencies in the supply chain elements and processes of [Assignment: organization-defined system or system component] in coordination with [Assignment:	Functional	Equal	Processes To Address Weaknesses or Deficiencies	TPM-03.3	Mechanisms exist to address identified weaknesses or deficiencies in the security of the supply chain	10	
SR-05	Acquisition Strategies, Tools, and Methods	Employ the following acquisition strategies, contract tools, and procurement methods to protect against, identify, and mitigate supply chain risks: [Assignment: organization-defined acquisition strategies, contract tools, and procurement	Functional	Intersects With	Acquisition Strategies, Tools & Methods	TPM-03.1	Mechanisms exist to utilize tailored acquisition strategies, contract tools and procurement methods for the purchase of Foreign Technology.	5	
SR-08	Notification Agreements	Establish agreements and procedures with entities involved in the supply chain for the system, system component, or system service for the [Selection (one or more): notification of supply chain compromises; results of assessments or audits:	Functional	Equal	Security Compromise Notification Agreements	TPM-05.1	Mechanisms exist to compel External Service Providers (ESPs) to provide notification of actual or potential compromises in the supply chain that	10	
SR-10	Inspection of Systems or Components	Inspect the following systems or system components [Selection (one or more): at random; at [Assignment: organization-defined frequency], upon [Assignment: organization-defined indications of need for inspection]] to	Functional	Intersects With	Product Tampering and Counterfeiting (PTC)	TDA-11	Mechanisms exist to maintain awareness of component authenticity by developing and implementing Product Tampering and	5	
SR-10	Inspection of Systems or Components	[Selection (one or more): at random; at [Assignment: organization-defined frequency], upon [Assignment: organization-defined indications of need for inspection]] to	Functional	Intersects With	Technology Asset Inspections	AST-15.1	Mechanisms exist to physically and logically inspect critical technology assets to detect evidence of tampering.	5	
SR-11	Component Authenticity	a. Develop and implement anti-counterfeit policy and procedures that include the means to detect and prevent counterfeit components from entering the system; andb. Report counterfeit system components to [Selection (one or	Functional	Intersects With	Product Tampering and Counterfeiting (PTC)	TDA-11	Mechanisms exist to maintain awareness of component authenticity by developing and implementing Product Tampering and	5	
SR-11(01)	Component Authenticity Anti-counterfeit Training	Train [Assignment: organization-defined personnel or roles] to detect counterfeit system components (including hardware, software, and firmware).	Functional	Equal	Anti-Counterfeit Training	TDA-11.1	Mechanisms exist to train personnel to detect counterfeit system components, including hardware, software and firmware.	10	
SR-11(02)	Component Authenticity Configuration Control for Component	Maintain configuration control over the following system components awaiting service or repair and serviced or repaired components awaiting return to service: [Assignment: organization-defined system components].	Functional	Equal	Maintain Configuration Control During Maintenance	MNT-07	Mechanisms exist to maintain proper physical security and configuration control over technology assets awaiting service or repair.	10	
SR-12	Component Disposal	Dispose of [Assignment: organization-defined data, documentation, tools, or system components] using the following techniques and methods: [Assignment: organization-defined techniques and methods].	Functional	Intersects With	Secure Disposal, Destruction or Re-Use of Equipment	AST-09	Mechanisms exist to securely dispose of, destroy or repurpose system components using organization-defined techniques and methods to	5	